

SISBID 2026 Module 5:

Deep Learning & Artificial Intelligence

Lecture 6:

Coding a Transformer from Scratch

Jean Feng + Ting Ye

What we'll do today

Let's code up our own simple transformer model!

This lab/lecture is based on Andrej Karpathy's minGPT GitHub repository, which is the pared down version of OpenAI's open-source GPT-2 implementation.



Transformer Decoder

GENERATING WIKIPEDIA BY SUMMARIZING LONG SEQUENCES

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GPT

Improving Language Understanding by Generative Pre-Training

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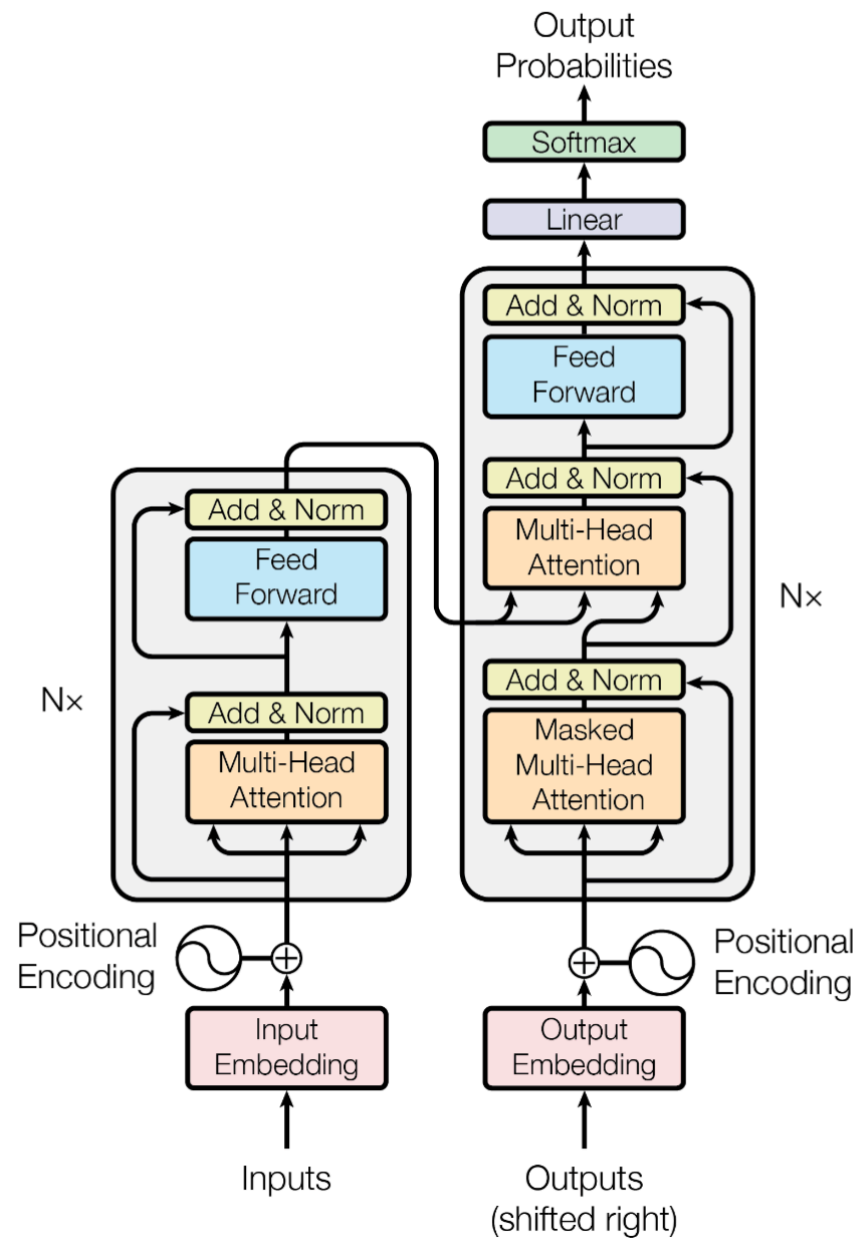
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GPT-2

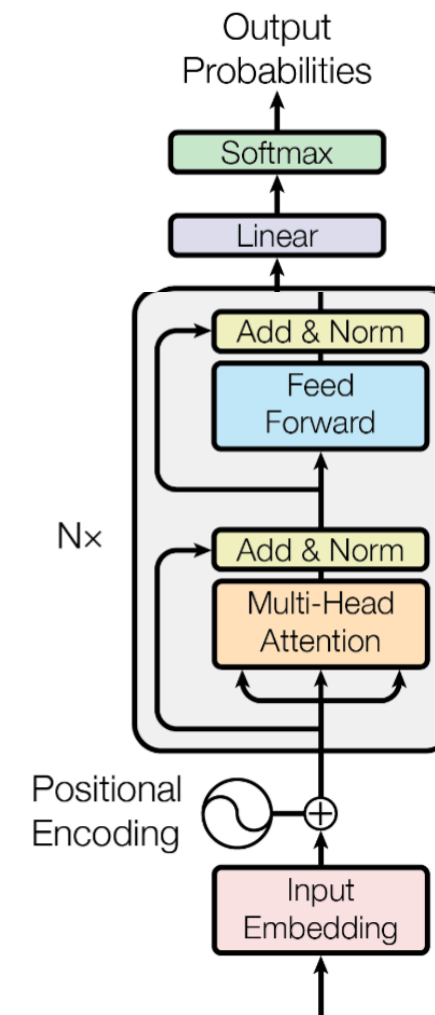
Language Models are Unsupervised Multitask Learners

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Transformer



Transformer Decoder



[Input tokens, Prior output tokens]

(Note: Masking applied so only look at previous words)

Coding up our own mini GPT-2

1. Go to: www.github.com/karpathy/minGPT
2. Clone the repo
3. [Optional] Setup a virtual environment so that you can run a simple demo.
4. We'll walk through all the code slowly together to understand what is going on under the hood.